# Discrete Mathematics and graph Theory
# Chapter Four:Directed graphs

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## Out lines

## Topics

In this chapter we will discuss the following topics:

**4.1.** Definition and examples of digraphs

**4.2.** Matrix representation of digraphs

**4.3.** Paths and connectivity

Directed graphs are graphs in which the edges are one way. Such graphs are frequently more useful in various dynamical systems such as:

- Digital computer.
- Flow system.
- Communication system.
- Transportation system.

## Definition

A digraph (directed graph) $D$ is a discrete structure defined as an ordered pair $D(V, E)$, where $V = V(D)$ is a non-empty set of vertices (or nodes or points) and $E = E(D)$ is a set of directed edges (or links/lines) connecting pairs of vertices.

- ☛ The arc from vertex $u$ to vertex $v$ is written as $(u, v)$

For a directed edge $e = (u, v)$, we use following terminologies.

- ✆ $u$ is the origin(initial node) and $v$ is the terminal (destination) of $e$.
- ✆ $v$ is the successor of $u$ and $u$ is predecessor of $v$.
- ✆ the arc is incident out of $u$ and incident into $v$.
- ✆ $u$ is adjacent to $v$ where as $v$ is adjacent from $u$.
- ✆ $u$ is a tail and $v$ is the head of $e$.

**Note:** If $u = v$ then $e = (u, v)$ is a loop.

## Remark

*The set of all successor of a vertex u is*

$$suss(u) = \{v \in V : (u, v) \in E\}$$

## Example

The graph $D$ below is a digraph. Find $succ(v_1)$, $succ(v_2)$ and $succ(v_3)$.

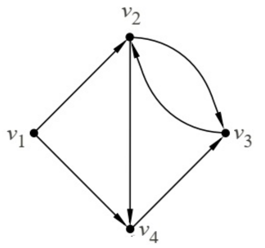



**Figure:** Pictorial representation of digraph D.

### Definition

If D is a digraph, then the undirected graph that results from ignoring the direction of the arcs in D is called the underlined graph of D

### Definition

Let $G(V, E)$ be a diagraph and let $V'$ be a subset of vertex set $V$. The graph $H(V', E')$ is said to be a sub diagraph of $G$ if $E'$ is a subset of $E$ such that the end points of the edges in $E'$ belongs to $V'$

### Example

A disconnected graph is the union of two or more connected subgraphs each pair of which has no vertex in common. These disjoint subgraphs are called the connected components of the graph.

## Degrees in the digraphs

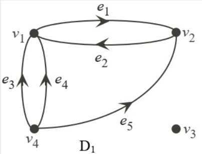
With directed graphs, the notion of degree splits into indegree and outdegree.

### Definition

Suppose $G$ is a direct graph. The out degree of a vertex $v$ of $G$, $outdeg(v)$ (0r $deg^{+}(v)$), is the number of edges incident from $v$, and the in degree of $v$, $indeg(v)$ (0r $deg^{-}(v)$), is the number of edges incident to $u$.

### Example

For the graph $D$ below, find in degrees and out degrees of the vertices.



**Theorem**

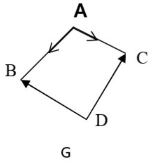
*The sum of the out degrees of the vertices of the diagraph $G$ equals the sum of the in degrees of the vertices, which equals the number of edges in $G$.*

**Note**

*A vertex $u$ in a diagraph with zero in degree is called a source and a vertex $u$ with zero outdegree is called a sink.*

**Example**

In the digraph $G$ below, determine the sources and sinks.



## Matrix Representation of a diagraph

### Definition

Suppose $G$ is a graph with $m$ vertices and suppose the vertices are ordered as $v_1, v_2, \cdots, v_m$. Then the adjacency matrix $A = (a_{ij})$ of the graph $G$ is the $m \times m$ matrix defined by:

$$a_{ij} = \begin{cases} n, & \text{if there are n edges from } v_i \text{ to } v_j \\ 0, & \text{otherwise,} \end{cases}$$

## Example

Determine the adjacency matrix for the graph $G$ below.

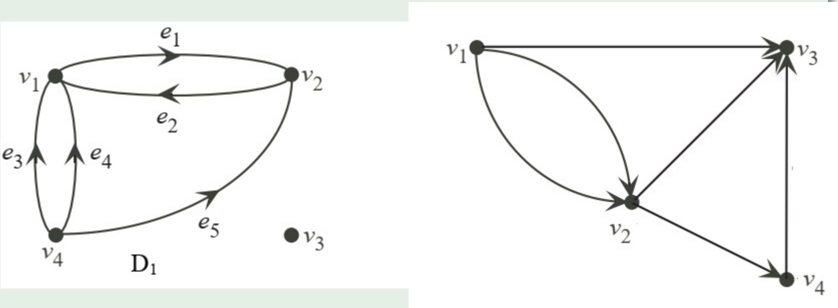



**Figure:** (a) Graph G, (b) Graph H

## Note

- ☞ *The sum of entries in the $j^{th}$ column is the same as* $indeg(v_j)$*.*
- ☞ *The sum of entries in the $i^{th}$ row is the same as* $outdeg(v_i)$*.*
- ☞ *Sum of entries in $A(D)$ is equal to the total number of arcs.*

## Exercise

*If matrix* $A(D)$ $\begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}$ *is the adjacency matrix of a diagraph $D$, then:*

**(a)** *Determine the in degree and out diagraph of each vertex.*

**(b)** *Determine the total number of edges of the graph $D$.*

**(c)** *Draw the digraph $D$.*

**(d)** *Find the component of $D$.*

## Incidence Matrix of a diagraph

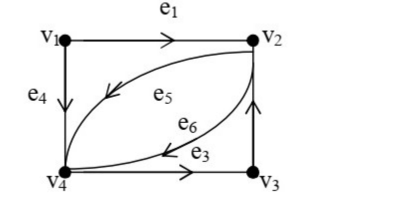
**Definition**

Suppose $D$ is a a loop-free diagraph with vertices $v_1, v_2, \cdots, v_m$. The incidence matrix of $D$ is $I(D) = (b_{ij})_{m\times n}$, where

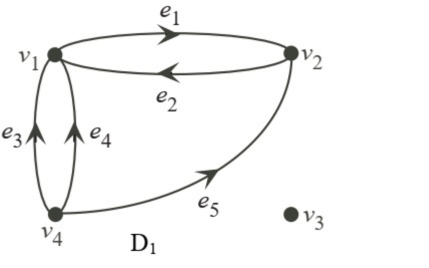
$$b_{ij} = \begin{cases} 1, & \text{if the arc } e_j \text{ is incident from } v_i \\ -1, & \text{if the arc } e_j \text{ is incident to } v_i \\ 0, & \text{otherwise,} \end{cases}$$

**Example**

Determine the incidence matrices for the following digraphs.




(a)




(b)

## Paths and Connectivity

The definitions for (directed) walks, paths, and cycles in a directed graph are similar to those for undirected graphs except that the direction of the edges need to be consistent with the order in which the walk is traversed.

### Definition

A directed graph $D(V, E)$ is said to be

- strongly connected if for every pair of nodes $u; v \in V$ , there is a directed path from $u$ to $v$ (and vice- versa) in $D$.
- unilaterally connected or unilateral(also called semi connected) if, for any pair of vertices $u$ and $v$, there is a path from $u$ to $v$ **OR** a path from $v$ to $u$ (one of them is reachable from the other).
- weakly connected or weak if replacing all of its directed edges with undirected edges produces a connected (undirected) graph.

## Example

Describe the connectivity( as strongly connected, unilaterally connected or weakly connected) for the following graph.

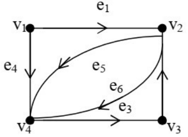



**Figure:** Digraph D.

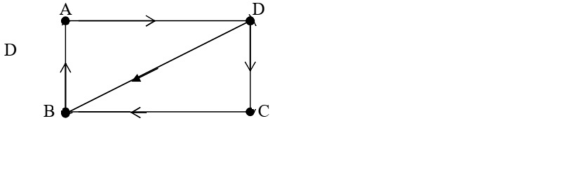
Answer: $D$ is not strongly connected (why?), It is unilaterally connected and it is weak as well.

## Definition

Spanning path of a directed(or non-directed graph) $G$ is a path that visits all the vertices of $G$.

## Example

Determine the spanning path of the following graph.



## Rooted Tree and Binary Tree

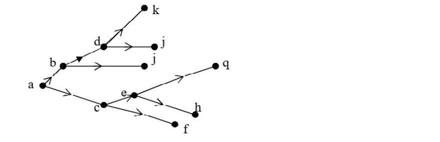
In the same manner as a family tree, we think of the root as the ancestor of all the other nodes in the tree, or equivalently, the other nodes are descendants of the root. A rooted tree can be defined formally as follows.

### Definition

A rooted tree is a connected diagraph $D$ with no cycles and with a unique vertex $v$ which has zero indegree. The vertex $v$ is called the root of the tree.

### Example

The following graph is a Directed tree rooted at vertex $a$.

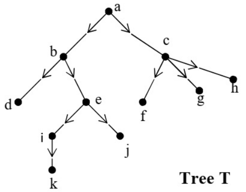


## Definition

- The length of the path from the root r to any vertex u is called the level of u.
- The depth of a node in the tree is the length (number of edges) of the (unique) path from the root to that node.
- Those vertices with zero outdegree are called leaves of the tree.
- The directed path from a vertex to a leaf is called a branch.

## Example

Consider the tree $T$ below.




**Tree T**

- vertex $a$ is the root of the tree. Vertex $a$ is at level 0. Vertices $b$ and $c$ are at level 1.
- Vertices $d$, $e$, $f$, $g$ and $h$ are at level 2.
- The vertices $d$, $k$, $j$, $f$, $g$, and $h$ with out degree zero are called leaves of the tree.
- Depth of $T$ is 4 which is level of $k$.

**Definition**

If there is a directed edge from a vertex $u$ to $v$, we say that $u$ is the parent of $v$ and that $v$ is the child of $u$.

- Nodes (vertices) that share the same parent are called siblings of each other.
- Vertices which have children are called internal vertices.

In the above graph, observe that $a$ is a parent of $d$ and $d$ is a child of $a$. $d\&e$, $b\&c$, $i\&j$ are siblings. $a$, $b$, $c$, $e$, $i$ are internal vertices.

**Definition (M-ary tree:)**

A rooted tree is called an $m - ary$ tree if every internal vertex has not more than $m-$children. The tree is called a full-$m-$ary if every internal vertex has exactly $m$ children.

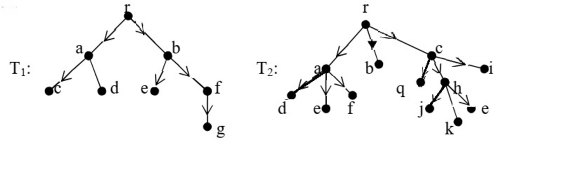
**Remark**

*An $m-$ary tree with $m = 2$ is called a Binary tree.*

In a binary tree, each node can have at most two children. A binary tree is either empty or consists of a node called the root together with two binary trees called the left subtree and the right subtree.

## Example

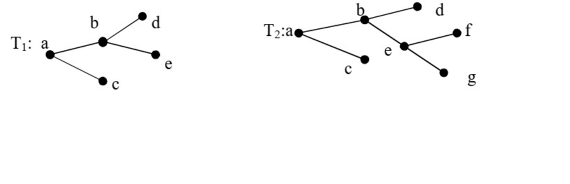
Consider the tree $T$ below.




☞ Notice that $T_1$ is a binary tree, $T_2$ is not a binary tree rather it is a full $3-$ary tree.

**Balanced tree:** A rooted $m - ary$ tree of depth $h$ is balanced if all the leaves are at level $h$ or $h - 1$.

**Example**




☞ Notice that $T_1$ is balanced binary tree of depth 2, $T_2$ is not balanced binary tree since leaf $c$ is neither at level 2 $(h - 1)$ nor at 3(or $h$).

**Binary search Tree:** is a binary tree in which each child is either a left or right child, no vertex has more than one left child and one right child, and the data associated with vertices must be unique.